

GREATER KANKAKEE, IL, USA

WMO: 722127

Lat: 41.072N

Lon: 87.847W

Elev: 192

StdP: 99.04

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.0	(c) -16.1	(d) -23.1	(e) 0.5	(f) -18.2	(g) -20.5	(h) 0.6	(i) -15.3	(j) 12.2	(k) -1.3	(l) 11.5	(m) -2.4	(n) 2.7	(o) 280	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.6	24.0	32.3	23.2	31.0	22.7	25.8	31.1	25.0	30.0	24.1	28.7	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.5	29.1	23.3	18.5	27.9	22.6	17.7	27.0	81.3	31.1	76.8	30.3	73.3	28.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.5	(c) 8.4	(e) -23.8	(f) 35.2	(g) 4.0	(h) 1.8	(i) -26.7	(j) 36.5	(k) -29.0	(l) 37.5	(m) -31.2	(n) 38.5	(o) -34.1	(p) 39.8		
(4) 11.1	(5) 9.5	(6) 8.4	(7) -23.9	(8) 27.6	(9) 3.7	(10) 0.9	(11) -26.6	(12) 28.2	(13) -28.7	(14) 28.8	(15) -30.8	(16) 29.3	(17) -33.5	(18) 29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-4.3	-2.5	4.3	10.5	16.7	21.7	23.6	22.4	19.3	12.3	5.1	-0.9
	DBStd	11.03	6.57	6.74	6.19	5.07	4.96	3.45	3.35	3.25	4.16	5.22	5.33	6.09
	HDD10.0	1587	443	351	198	54	4	0	0	0	0	34	162	340
	HDD18.3	3298	701	583	437	238	91	9	2	6	37	202	396	597
	CDD10.0	1868	1	2	22	69	213	352	421	385	281	104	17	2
	CDD18.3	535	0	0	2	3	42	112	164	133	67	13	0	0
	CDH23.3	5214	0	0	16	55	456	1034	1661	1222	637	132	1	0
	CDH26.7	1879	0	0	2	4	152	360	664	435	223	38	0	0
Wind	WSAvg	3.6	4.3	4.2	4.3	4.6	3.8	3.1	2.5	2.1	2.5	3.5	4.0	4.0
Precipitation	PrecAvg	951	58	43	59	91	110	98	122	102	68	77	72	54
	PrecMax	1297	126	125	109	182	177	244	279	238	201	182	152	110
	PrecMin	774	13	11	22	38	14	0	53	34	10	15	3	8
	PrecStd	139	33	30	28	38	49	57	63	59	52	44	39	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	17.2	26.0	27.4	32.8	34.1	36.2	34.1	33.9	31.1	21.9	16.3
		MCWB	13.9	13.1	17.7	17.1	21.5	23.0	26.2	24.1	22.5	21.8	14.9	14.0
	2%	DB	10.2	12.9	20.8	25.0	30.8	32.4	33.8	32.6	31.9	27.1	18.0	12.8
		MCWB	9.0	10.0	14.7	16.0	20.7	22.9	25.2	23.7	22.5	18.2	13.8	10.8
	5%	DB	7.1	8.8	17.3	22.6	28.0	30.8	32.3	31.1	29.1	23.8	16.1	10.0
		MCWB	5.5	6.8	12.9	14.8	19.0	22.4	24.1	23.1	21.2	17.5	12.4	8.6
	10%	DB	4.0	6.0	13.9	20.0	26.0	28.6	30.5	28.9	27.3	21.2	13.3	7.3
		MCWB	2.8	4.0	10.5	13.7	18.4	21.3	23.3	22.3	19.9	16.6	10.1	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	14.6	18.0	19.3	23.5	25.8	27.7	26.7	25.1	22.8	16.8	14.8
		MCDB	15.5	16.1	24.2	24.1	29.7	30.9	33.6	31.7	31.8	28.6	19.3	16.0
	2%	WB	9.2	10.5	15.8	17.5	21.9	24.7	26.4	25.4	23.7	20.8	14.9	11.6
		MCDB	9.8	12.7	19.9	22.5	27.9	29.7	31.9	30.3	29.1	24.8	17.7	12.6
	5%	WB	5.5	7.0	13.8	15.8	20.7	23.6	25.4	24.4	22.6	19.1	12.8	8.4
		MCDB	7.2	8.5	16.5	20.9	25.8	28.3	30.5	28.8	27.0	21.6	15.4	9.5
	10%	WB	2.9	3.9	10.8	14.5	19.5	22.6	24.4	23.5	21.5	16.9	10.7	5.8
		MCDB	4.0	5.4	13.9	19.4	24.3	27.2	29.0	27.5	25.4	20.5	13.3	7.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	10.1	12.0	12.1	11.3	11.3	11.6	12.4	11.2	9.4	7.4
		MCDBR	10.4	12.4	14.1	16.0	14.9	13.2	12.5	13.1	14.2	14.6	12.8	10.0
		MCWBR	9.2	10.0	9.3	9.1	7.0	5.8	5.7	5.8	6.0	7.4	8.9	8.9
	5% WB	MCDBR	10.2	12.0	12.6	14.1	12.5	11.1	11.3	10.7	11.7	11.4	11.0	9.4
		MCWBR	9.3	10.3	9.0	9.1	6.8	5.9	6.0	5.7	5.8	7.0	8.6	8.8
Clear-Sky Solar Irradiance	taub	0.296	0.323	0.352	0.405	0.440	0.441	0.460	0.452	0.421	0.361	0.318	0.296	
	taud	2.426	2.344	2.359	2.221	2.177	2.218	2.157	2.168	2.227	2.397	2.469	2.450	
	Ebn at Noon	862	884	896	866	841	838	817	812	811	830	828	831	
	Edh at Noon	79	100	111	136	146	141	148	142	125	94	76	72	
All-Sky Solar Radiation	RadAvg	1.70	2.58	3.62	4.67	5.33	6.11	6.10	5.37	4.50	3.01	1.96	1.42	
	RadStd	0.18	0.30	0.27	0.38	0.45	0.39	0.31	0.31	0.36	0.32	0.19	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page